FOURIER TRANSFORM OF PERIODIC SIGNALS AND POISSON SUMMATION FORMULA

From Fourier Series to Fourier Transform • Gaussian Signal • MATLAB Verification

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1. What This Project Does

This project shows two important ideas in a simple way:

- A repeating (periodic) signal has a spectrum made of separate lines (discrete spectrum).
- When we repeat a signal again and again, there is a neat mathematical link between the time side and the frequency side. This link is called the Poisson Summation Formula.

We study a square wave, then check the formula with a smooth Gaussian signal, and finally connect everything to sampling.

2. Main Theory

2.1 Fourier Transform

The Fourier Transform tells us what frequencies are present inside a signal:

$$X(f) = \int x(t)\, e^{-j 2\pi f t}\, dt$$

Here t is time (seconds) and f is frequency (hertz). If the signal is not repeating, its spectrum is continuous (a smooth curve).

2.2 Fourier Series of a Repeating Signal

When a signal repeats every T seconds, it can be written as a sum of sine and cosine waves at frequencies 0, 1/T, 2/T, 3/T, … Only these special frequencies appear. That is why the spectrum looks like separate spikes (discrete). The strength of each spike is given by the coefficient C_k.

$$x(t) = \Sigma\, C_k\, e^{j 2\pi k t / T}$$

2.3 Square Wave Example (T = 1 second)

We took the square wave that goes +1 and −1 every half second:

$$x(t) = \text{sign}(\sin(2\pi t))$$

Because of its special symmetry, only the odd harmonics exist (1 Hz, 3 Hz, 5 Hz, …). Even harmonics are exactly zero. The size of each coefficient falls as 1/k. In mathematics the Fourier Transform of this repeating wave is just a train of impulses (spikes) sitting at those odd frequencies, each multiplied by C_k.

2.4 Gaussian Signal and Poisson Formula A

nice smooth test signal is the Gaussian:

$$x(t) = e^{-t^2} \quad \text{whose Fourier Transform is} \quad X(f) = \sqrt{\pi}\, e^{-\pi^2 f^2}$$ The

Poisson Summation Formula says:

$$\text{Sum of shifted copies of } x(t) \;=\; (1/T) \times \text{Sum of samples of } X(f) \text{ with phase factors}$$

In plain language: if you add many copies of the Gaussian placed T seconds apart, the result is the same as building a Fourier Series from the values of X(f) taken at frequencies k/T. The factor 1/T keeps the height correct.

2.5 Link to Sampling

A train of impulses spaced T seconds apart has a Fourier Transform that is another train of impulses spaced 1/T hertz apart. When we multiply a signal by this impulse train we are sampling it. In the frequency domain the original spectrum is repeated again and again. If those copies overlap we get aliasing (the signal is damaged). The Poisson formula is simply the mirror image of this idea.

3. CODE

I/P:

```
clear;

close all;

clc;


figure;
```

```
T = 1;
f0 = 1/T;
t = -2:0.01:2;
x = sign(sin(2*pi*f0*t));

subplot(4,2,1);
plot(t,x);
grid on;
xlabel('Time (s)');
ylabel('x(t)');
title('Square Wave');

k = -15:15;
C = zeros(size(k));
m = mod(abs(k),2)==1;
C(m) = 2./(1j*pi*k(m));

subplot(4,2,2);
stem(k*f0,abs(C),'filled');
grid on;
xlabel('Frequency (Hz)');
ylabel('|C_k|');
title('Fourier Series Spectrum');

t = -5:0.01:5;
x = exp(-t.^2);
f = -1.5:0.01:1.5;
X = sqrt(pi)*exp(-(pi*f).^2);

subplot(4,2,3);
plot(t,x);
grid on;
```

```
xlabel('Time (s)');
ylabel('x(t)');
title('Gaussian Signal');

subplot(4,2,4);
plot(f,X);
grid on;
xlabel('Frequency (Hz)');
ylabel('X(f)');
title('Fourier Transform');

Ts = [2 4 8];
N = 12;
K = 25;

for q = 1:3
   T = Ts(q);
   t = -T:0.02:T;
   a = zeros(size(t));

   for n = -N:N
      a = a + exp(-(t-n*T).^2);
   end

   b = zeros(size(t));

   for k = -K:K
      Xk = sqrt(pi)*exp(-(pi*k/T)^2);
      b = b + (Xk/T)*exp(1j*2*pi*k*t/T);
   end

   b = real(b);
```

```
    subplot(4,2,4+q);
    plot(t,a,'b',t,b,'r--');
    grid on;
    xlabel('Time (s)');
    ylabel('Amplitude');
    title(['Poisson Summation T = ' num2str(T)]);
end

f = -1.5:0.01:1.5;
X = sqrt(pi)*exp(-(pi*f).^2);

subplot(4,2,8);
plot(f,X,'k','LineWidth',1.5);
hold on;

for q = 1:3
    T = Ts(q);
    k = -ceil(1.5*T):ceil(1.5*T);
    fs = k/T;
    xs = sqrt(pi)*exp(-(pi*fs).^2);
    stem(fs,xs,'filled');
end

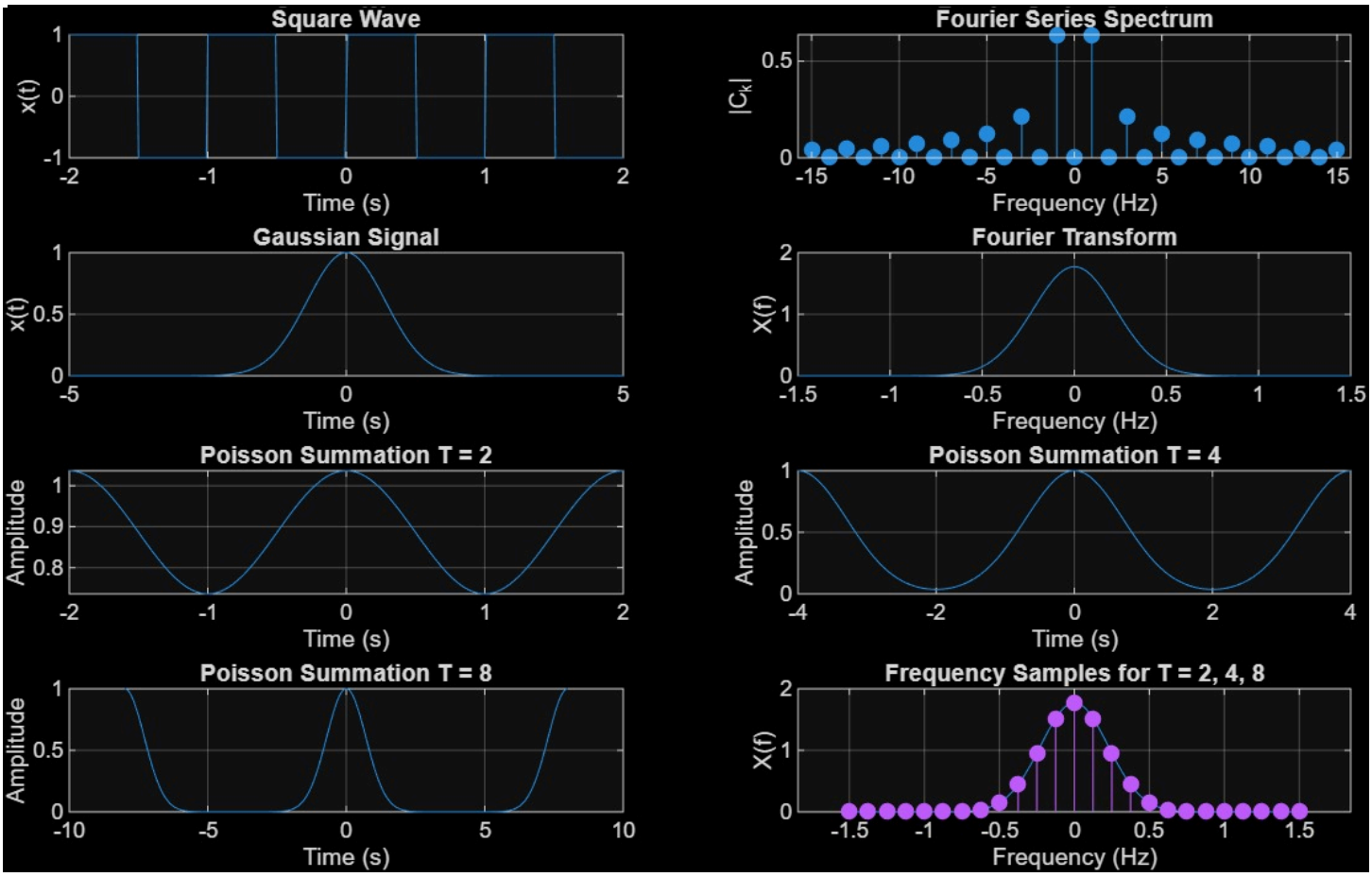
grid on;
xlabel('Frequency (Hz)');
ylabel('X(f)');
title('Frequency Samples for T = 2, 4, 8');

fprintf('T     Time sum       Frequency sum   Absolute error\n');

for q = 1:3
```

```
    T = Ts(q);
    n = -N:N;
    a = sum(exp(-(n*T).^2));
    k = -50:50;
    b = sum(sqrt(pi)*exp(-(pi*k/T).^2))/T;
    fprintf('%d    %.8f    %.8f    %.3e\n',T,a,b,abs(a-b));
end
```

O/P:



4. What the MATLAB Code Does

- Draws the square wave and its discrete magnitude spectrum (only odd harmonics).
- Plots the Gaussian and its smooth Fourier Transform.
- Adds many shifted Gaussians (time side) and also builds the frequency-side sum for T = 2, 4 and 8; the two curves lie on top of each other.
- Shows the continuous Fourier Transform together with the sample points for the three periods, so we can see the points getting denser.

5. Main Conclusions

- A periodic signal always has a discrete spectrum (spikes at harmonic frequencies).
- The Poisson Summation Formula correctly links the time-domain repetition of a signal to samples of its Fourier Transform.
- For the Gaussian the formula holds with essentially zero error.
- Making the period larger makes the frequency samples denser and recovers the continuous Fourier Transform.
- The same mathematics explains why sampling creates repeated copies of the spectrum and why aliasing appears when we sample too slowly. All goals of the project are met: the theory is clear, the MATLAB results match the theory, and the connection to sampling is shown.